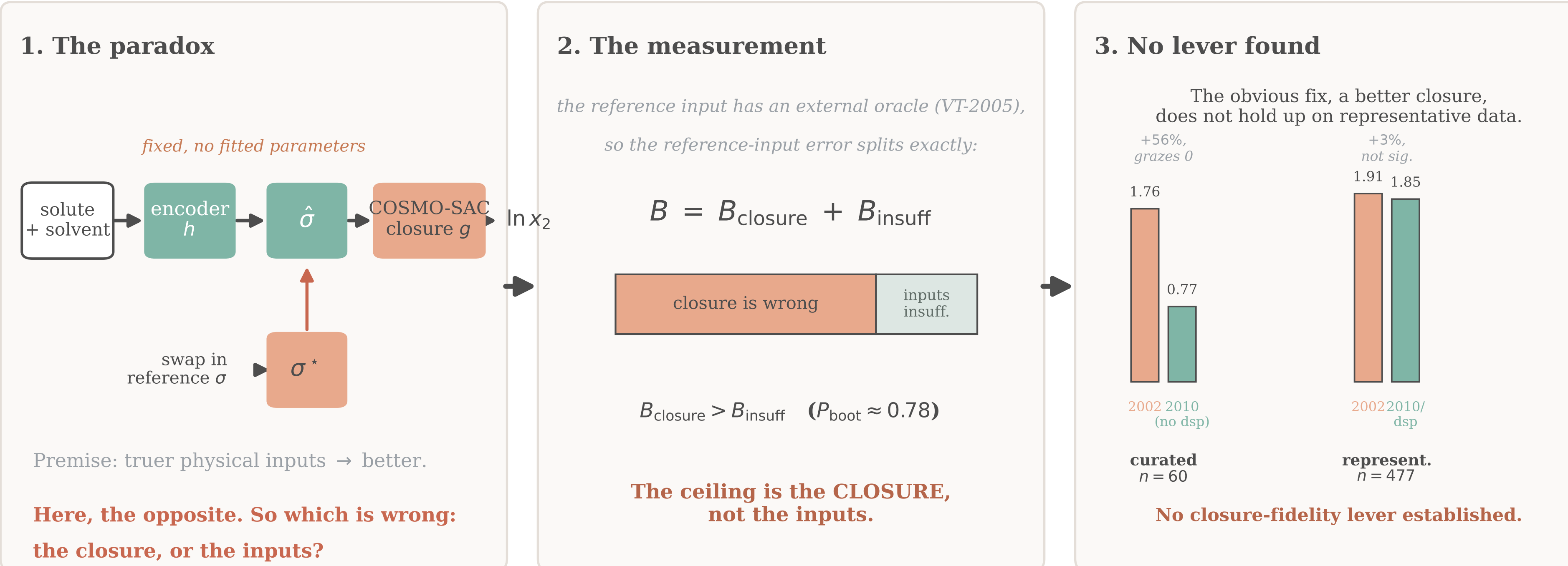
When do reference physical inputs help a learned solubility model?



What the paper establishes, graded by evidence

CERTIFIED

- Reference σ makes solubility worse (3 seeds, $n = 5608$)
- Exact split $B = B_{\text{closure}} + B_{\text{insuff}}$ (Lemma 2)
- pKa flip: helps meta/para, hurts ortho ($\approx 20/20$)

LIKELY

- Latent is a low-rank transferable surrogate ($n = 44$)
- Closure is the larger term (low-γ, $P_{\text{boot}} \approx 0.78$)
- Attribution holds on the matched $n = 60$ corner

OPEN

- No fidelity lever: 2010/dsp +3% ($n = 477$, $P = 0.62$)
- Transfer of the attribution to finite composition
- Whether the drift is the closure's inverse $J^{+}(m - g)$